Q FOR EMAIL: OENG1209-Q / PROBLEM/ISSUE

OENG1209: Engineering Science

Week 1:

Course Introduction + Motion

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Tentative Outline

- Week 1 – 4: Kinematic and Dynamic systems.
- Week 5 – 6: Conservation of Momentum and Energy.
- Week 7 – 8: Electric Field and Magnetic Fields.
- Week 9 -10: Oscillation and Waves.
- Week 11: Review.

How to succeed in the class

- Show up and participate in class lecture/tutorial/labs.
- Having a solution is good, having a strategy is better.
- The only stupid question is the one that was not asked.
- Personal conversations should be taken outside of classroom.
- Make sure your work is your own and be proud of your own work.
- No form of plagiarism will be tolerated.

POST: SUMARISE WHAT U LEARN

ASSESSMENTS IS OPEN BOOKS

Today's Topics

- **Physical Units and Problem Solving in Physics.**
- **Vector Introduction.**
- **Vector Arithmetic.**
- **The Parameters of Motion.**

Physical units and Problem solving in Physics

The laws of nature and how to quantify them

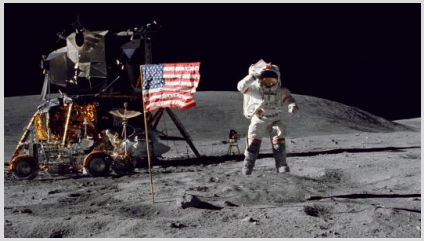
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Introduction to Engineering Physics

- **Physics** is the study of the fundamental laws of nature that underlie all physical phenomena in the universe.







- Remarkably, we have found that these laws can be expressed in terms of **mathematical equations** and as a result, it is possible to make precise, quantitative comparisons between the predictions of theory and the observations of experiments.

Physical units

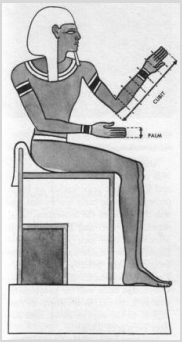
- To make **quantitative comparisons** between the laws of physics and our observation of the natural world, we must define some measurable physical quantities such as: **Length (L)**, **Mass (m)**, **Time (t)**,
- In the mainstream scientific community, the **SI (Système international)** system is often considered the official units of physical measurements.

These are mechanics Quantities



Physical units – Length

- Early **units of length** were often associated with the human body.
  - The Egyptians used a unit call **"cubit"** which is the distance from the elbow to the tip of the middle finger.
  - The English **foot** used to be the length of the foot of King Louis XIV.
- The official SI unit of length is a **meter**, which is currently defined as "the distance traveled by light in a vacuum in 1/299,792,458 second.







Physical units – Mass

- In East Asia, the traditional unit of mass is **Tael** (兩, lượng, 량) which originated from China.
- In SI units, mass is measured in **kilograms (kg)** which used to be defined as the mass of a particular platinum – iridium alloy cylinder at the International Bureau of Weights and Measures in Sevres, France.
- From May 20th 2019, the kilogram is defined base on the **Planck constant**.



1 Tael = 0.0378 kg



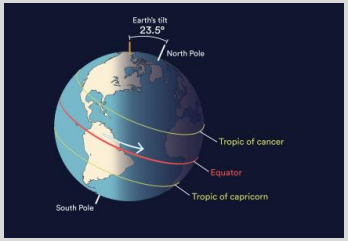
The standard kilogram.

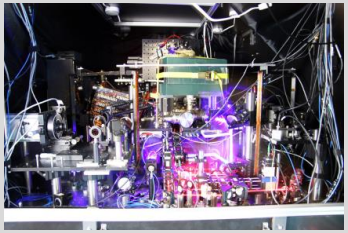
Physical units – Time

- Virtually all countries and cultures in the world measured **time** based on the rotation of the Earth.

$$1\ Earth\ rotation = 1\ day = 24\ hours = 86400\ seconds$$

- Today, we use advance timekeeping equipment called **the "atomic clocks"** which are based on characteristic frequencies of radiation emitted by **Cesium-133 isotope** atoms.
- The current definition of a second is the time it takes for "radiation from a Cesium-133 atom to complete 9,192,631,770 cycles of oscillation".





Physical units – Dimensional Analysis

- In physics, when we refer to **the dimension** of a physical quantity, we are referring to **the type of quantity** in question, regardless of the unit used.
  - Example: a cubit and a lightyear are both in the same dimension of length.
- For a formula in physics to be valid, it must be dimensionally consistent, which means each term in the equation must have the same dimensions.



This statement is **dimensionally inconsistent** as parsec is a unit of length, not time.

Physical units – Dimensional Analysis

- To check the **dimensional consistency** of an equation, it is convenient to use **a square bracket** [ ] to indicate the dimension of a quantity.
- If x represents a distance, which has dimensions of **length** $[L]$**,** we write:

$$x = [L]$$

- If v represents a velocity which has a dimensions of length $[L]$ per time $[T]$, we write:

$$v = \frac{[L]}{[T]}$$

Physical units – Dimensional Analysis

- The operation of checking Dimensional Consistency is also called **Dimensional Analysis**. Let's perform dimensional analysis on this equation:

$$x = x_0 + vt$$

- Here it is given that x and x_0 are distances, v is velocity, and t is time. Thus we can write the dimensions of each terms as:

$$[L] = [L] + \frac{[L]}{[T]}[T] = [L] + [L]$$

- We can see that each term in this formula has the same dimension, $[L]$ which means the equation is **dimensionally consistent**.

Physical units – Dimensional Analysis Example

- **Example:** The time T required for one complete oscillation of a mass m on a spring with spring constant k is given by the equation:

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Find the dimensions of the spring constant k in order to allow the equation to be dimensionally correct.

$T = 2\pi\sqrt{m/k}$ $\quad$ $k = m/(T/2\pi)^2$

$\frac{T}{2\pi} = \sqrt{m/k}$ $\quad$ $k = [m]/[T]$

$\left(\frac{T}{2\pi}\right)^2 = m/k$

Physical units – Converting Units

- It's often convenient to convert from one set of units to another. For example:

$$1\ m = 3.281\ ft$$

$$\rightarrow \frac{\mathbf{1}\boldsymbol{m}}{\mathbf{3.281}\boldsymbol{ft}} = \mathbf{1}$$

- If you meet someone who said they ran **316 $\boldsymbol{ft}$** just now, how should we convert this distance to meters?

$$(\mathbf{316}\ \boldsymbol{ft})\left(\frac{\mathbf{1}\boldsymbol{m}}{\mathbf{3.281}\boldsymbol{ft}}\right) = \mathbf{96.3}\ \boldsymbol{m}$$

- If you want to convert more than one unit at a time, you can keep adding converting factors to the equation in such a way that the unwanted units are cancelled leaving only the desirable units.

Physical units – Converting Units

- **Example:** The interior of a popular type of microwave oven has a width of $W = 15.5\ imches$, a depth of $D = 14.5\ inches$, and a height of $H = 9.25\ inches$. What is the interior volume of the oven in SI units?

$(15.5)(14.5)(9.25)$

$(0.0254)^3$



Solving Problems in Physics

- **Physics** and other **scientific and engineering fields** are like physical sports, you have to learn by doing.
- Reading and memorizing formula is good but actually forming **a problem solving strategy** to handle different kinds of problems is better.
- Some Problem-solving suggestions:

PROBLEM

1. Read the problem carefully.
2. Sketch the system.
3. Visualize the physical process. (TABLE PHY QUANTITY)
4. Strategize
5. Identify appropriate equations
6. Solve the equations
7. Check your answer.
8. Explore limits/special cases

Solving Problems in Physics – Exercise

- **Exercise:** Male mosquitoes in the mood for mating find female mosquitoes of their own species by listening for the characteristic “buzzing” frequency of the female’s wing beats. This frequency is about 605 wing beats per second. A student would like to know:
  - How many wing beats does the female mosquito make in one minute?
  - How many cycles of oscillation does the radiation from a cesium-133 atom complete during one mosquito wing beat?



Vector Introduction

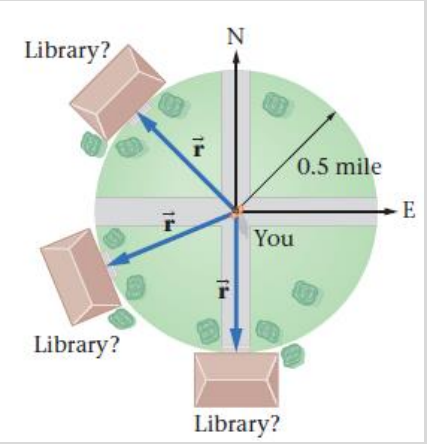
What is it and what make them so fun?

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Vector Definition

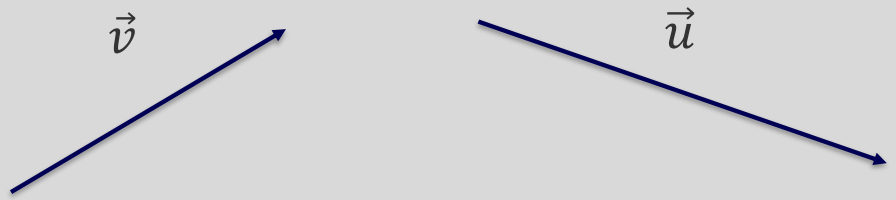
- You have just visited RMIT for the first time. As you exit the parking lot, you ask a fellow student “Do you know where the library is?”. He replied: “Yes, it is half a mile away from here”. How helpful is that?





Vector Definition

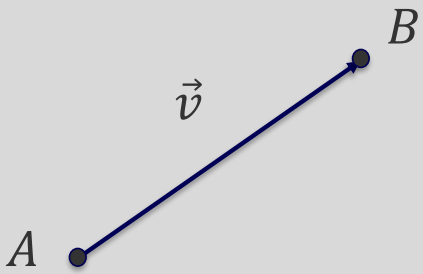
- A **scalar** is a numerical value, expressed in terms of appropriate units.
- A **vector** is a physical quantity with both a numerical value (magnitude) and a direction.
- Vectors are often represented by an arrow or a line segment with direction.
  - Example:



Vector Notations

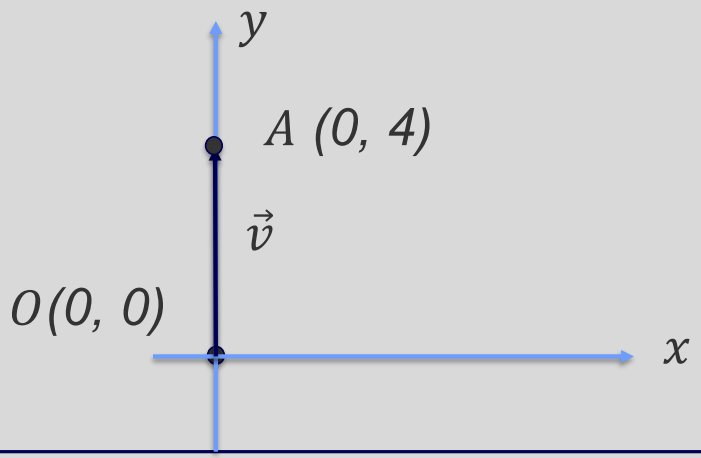
- **Vectors names** are usually accented with an arrow pointing to the right side. Example: $\vec{v}$, $\vec{u}$, $\vec{k}$, $\overrightarrow{AB}$, $\overrightarrow{CD}$, *etc.*

- **Example:** vector $\vec{v}$ is defined as the line from point A to point B. Thus, we can refer to it as vector $\vec{v}$ or vector $\overrightarrow{AB}$.



Vector Components

- **Magnitude** is the length, or size, of a vector.
- **Direction** is to indicate the orientation of the vector with respect to some pre-defined coordinate system.
- **Example:**

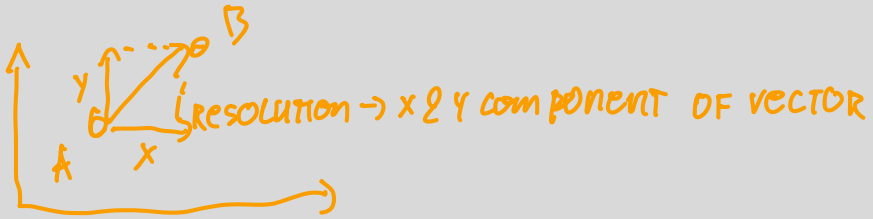


Vector $\overrightarrow{OA}$ has the length of 4 and pointing at 90° angle from the x-axis.

Vector Components

- The process of splitting a vector up into its x, y, z components is called **vector resolution**.
- Vector resolution shall be indicated as a pair of number separated with a comma within a pair of square bracket [..].
- For any vector $\vec{v}$ between point A with coordinate (x_1, y_1) and point B with coordinate (x_2, y_2), the resolution of vector $\vec{v}$ shall be equal to:

$$[x_2 - x_1, y_2 - y_1].$$

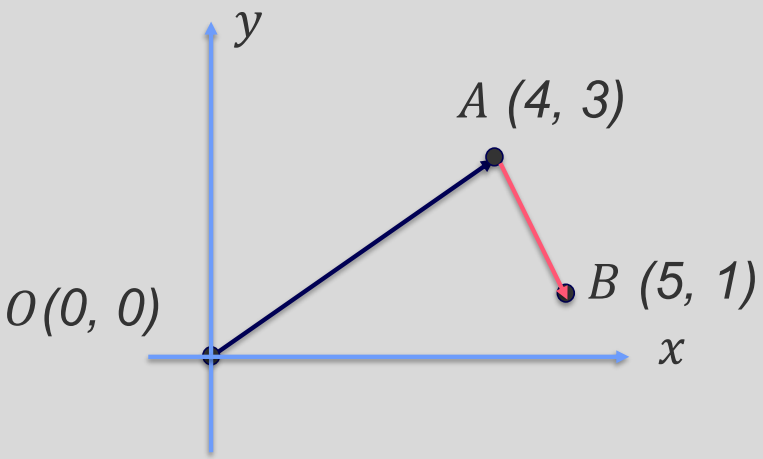


Vector Components

- **Example:** Find the resolutions of Vector $\overrightarrow{OA}$ which goes from point O (0,0) to point A (4, 3), and Vector $\overrightarrow{AB}$ which goes from point A (4,3) to point B (5, 1).

$\overrightarrow{OA} = [4-0, 3-0] = (4, 3)$

$\overrightarrow{AB} = [5-4, 1-3] = (1, -2)$



Vector Components - Magnitude

- The magnitude of a Vector is denoted as the name of the vector with **Vertical Bars on each side |…|**.
  - Example: the magnitude of vector $\overrightarrow{AB}$ is written as $|\overrightarrow{AB}|$.
- To calculate the magnitude of a Vector, we invoke the **Pythagorean Theorem.**
- Thus, for any vector $\vec{V}$ with resolution [x, y], the magnitude of $\vec{v}$ shall be

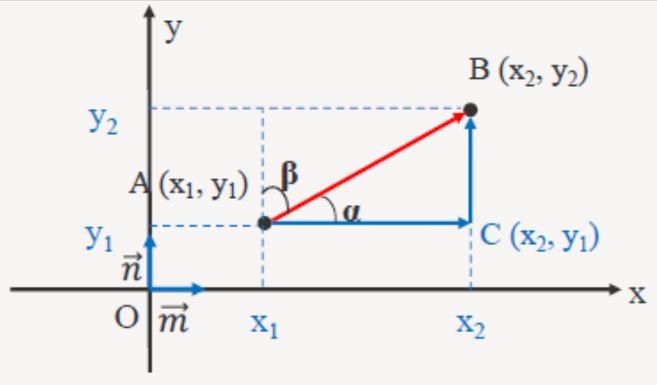
$$|\vec{v}| = \sqrt{x^2 + y^2}$$

VECTOR CAN BE WRITE AS $\underset{\sim}{V}$



Vector Components - Direction

- The direction of a vector is usually expressed by a quantity called **"direction cosine"**.



$$\cos\alpha = \frac{(x_2 - x_1)}{|\overrightarrow{AB}|}$$

$$\cos\beta = \frac{(y_2 - y_1)}{|\overrightarrow{AB}|}$$

For 2D Vectors, α and β are the angles the vector makes with the x-axis and y-axis respectively.

$$\boldsymbol{\alpha = \cos^{-1}\left(\frac{(x_2 - x_1)}{|\overrightarrow{AB}|}\right)}$$

$$\boldsymbol{\beta = \cos^{-1}\left(\frac{(y_2 - y_1)}{|\overrightarrow{AB}|}\right) = \sin^{-1}\left(\frac{(x_2 - x_1)}{|\overrightarrow{AB}|}\right)}$$

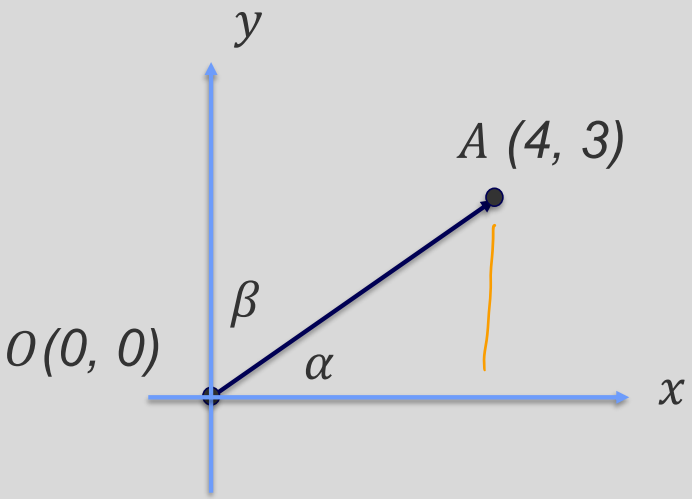
Vector Components - Magnitude

- **Example:** Calculate the magnitude and direction angles α and β of vector $\overrightarrow{OA}$ [4, 3].

$$|\overrightarrow{OA}| = \sqrt{4^2 + 3^2}$$

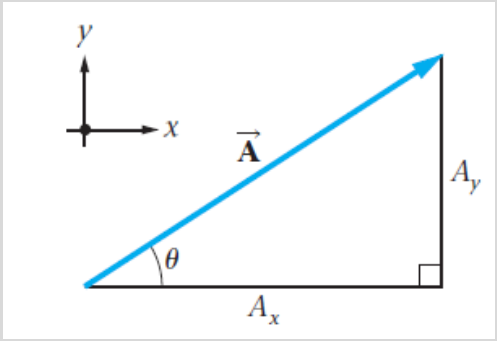
$$\alpha = \cos^{-1}\left(\frac{4-0}{|\overrightarrow{OA}|}\right)$$

$$\beta = \cos^{-1}\left(\frac{3-0}{|\overrightarrow{OA}|}\right)$$



Vector Components - Conversion

- Sometimes, we are given **the Magnitude** $|\vec{A}|$ and **Direction angle** (θ) of a vector and is required to find the vector's x and y components.
- This can be done by invoking the trigonometric relationships of the sides of a right triangle, as any 2D vector can be approximated by **a right triangle** whose sides are its x and y components.



Given a vector $\vec{A}$ with magnitude A and direction angle θ. Its components A_x and A_y can be calculated as:

$$A_x = |\vec{A}|cos\theta$$
$$A_y = |\vec{A}|sin\theta$$

$\frac{A_x}{|\vec{A}|} = cos\theta$ $\frac{A_y}{|\vec{A}|} = sin\theta$

$A_x = cos\theta\,(|\vec{A}|)$

Vector Components - Conversion

- **Example:** The vector $\vec{A}$ has a magnitude of 9.75 meters. Find its components if its direction angles (measured from the $x-axis$) are:
  - $\theta_1 = 7.25^o$
  - $\theta_2 = 115^o$
  - $\theta_3 = 235^o$
  - $\theta_4 = 355^o$

$x = \cos(7.25)\ 9.75$

VECTOR $\vec{A}$: 9m $(\alpha = 60^\circ)$



$A_x = |\vec{A}|\cos(\alpha)$

$A_y = |$

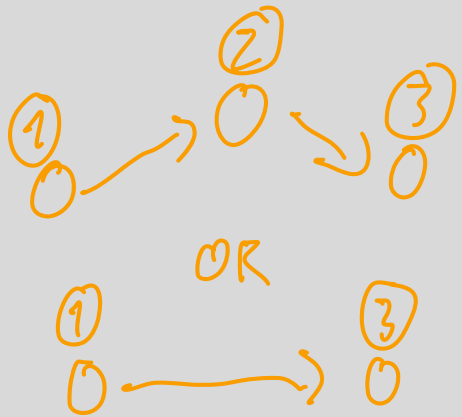
Vector Arithmetic

All the fun things we can do with Vectors.

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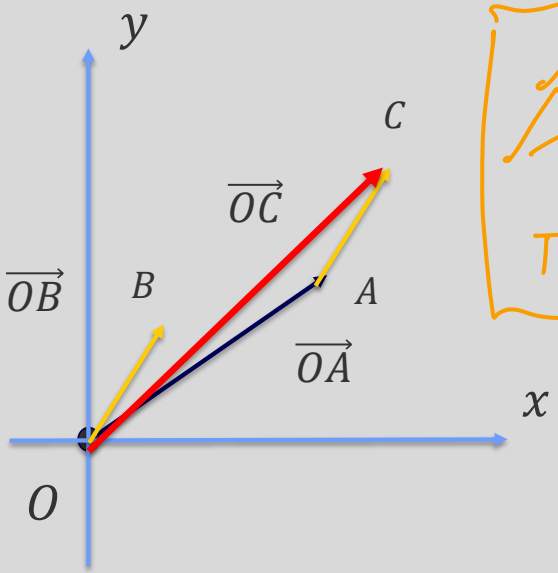
Vector Arithmetic – Overview

- In order to solve problems with vectors, it is helpful to understand what happens when we **add, subtract, divide, and multiply** with vectors.
- Vector arithmetic operations will result in another vector or a scalar (only for multiplication by dot product).



Vector Arithmetic – Addition

- Vector addition is when we try to add one vector to another vector. The result will be another vector.
- When we add a vector $\overrightarrow{OB}$ to vector $\overrightarrow{OA}$, the result will be vector $\overrightarrow{OC}$.



ADDING RULE
TRIANGLE RULE

PARALLEL RULE

Vector Arithmetic – Addition

- The resolution [x_r, y_r,] of the resultant vector $\overrightarrow{OC}$ can be calculated from the resolution [x_1, y_1] of vector $\overrightarrow{OA}$ and the resolution [x_2, y_2] of vector $\overrightarrow{OB}$ as follow:

$$\boldsymbol{\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}}$$

$$\rightarrow [x_1, y_1,] + [x_2, y_2] = [x_r, y_r]$$

$$\rightarrow \boldsymbol{[x_r, y_r] = [x_1 + x_2, y_1 + y_2]}$$

Vector Arithmetic – Subtraction

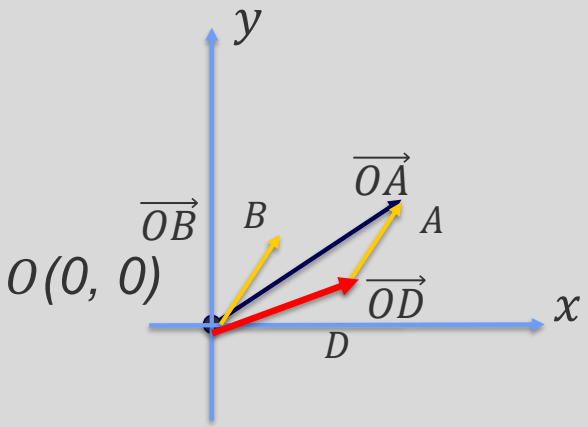
- Subtracting one vector from another vector is the same as adding the **negative vector** of one vector to another vector, i.e. 1 - 1 is the same as 1 + (-1).
- A **negative vector** is a vector that has the **same magnitude** as the original vector but has its **direction flipped 180 degrees** around the x, y, and z-axis. All x, y, z components of that vector will have their signs switched from positive to negative and vice versa.
  - **Example:** The negative of vector $\vec{u}[3, 4]$ is vector $-\vec{u}[-3, -4]$

$\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$



Vector Arithmetic – Subtraction

- In 2D, **vector subtraction** can be illustrated as follow:



- The resolution [x_r, y_r] of the resultant vector $\overrightarrow{OD}$ can be calculated as follow:

$$\overrightarrow{OA} - \overrightarrow{OB} = \overrightarrow{OD}$$

$$\rightarrow [x_1, y_1] - [x_2, y_2] = [x_r, y_r]$$

$$\rightarrow [x_r, y_r] = [x_1 - x_2, y_1 - y_2]$$

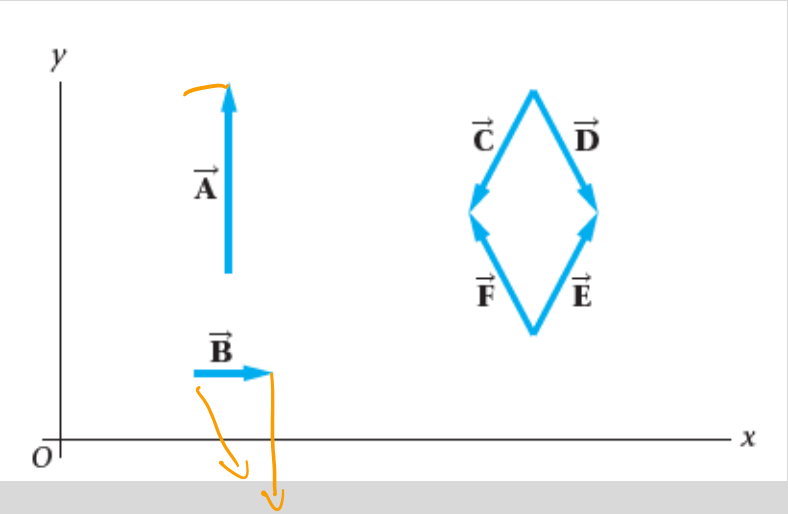
Vector Arithmetic – Example 1

- **Example:** Consider the vectors $\vec{A}$ and $\vec{B}$ shown in the figure below. Which of the other four vectors in the figure ($\vec{C}, \vec{D}, \vec{E}, and\ \vec{F}$) best represents the direction of

  a) $\vec{A} + \vec{B}$ → E

  b) $\vec{A} - \vec{B}$ → F

  c) $\vec{B} - \vec{A}$ → D



Vector Arithmetic – Scalar Multiplication

- A vector $\vec{u}$ can be multiplied or divided by **a scalar "k"** to **yield another vector $\overrightarrow{u_1}$.**
- The resolution of $\overrightarrow{u_1}$ can be calculated as follow:

$$\overrightarrow{u_1} = k * \vec{u}$$

$$\rightarrow \overrightarrow{u_1}[x_1, y_1] = k * \vec{u}[x, y]$$

$$\rightarrow \overrightarrow{u_1}[x_1, y_1] = [k * x, k * y]$$

Vector Components – Unit vector

- **Unit vector** is a vector which **has a magnitude of 1**. Unit vectors are often denoted with an **accented "^"**.
- Example:
  - Is Vector $\vec{a}$ [0, 1] a unit vector? yes
  - Is Vector $\vec{b}$ [1, 1] a unit vector? no
- To find the unit vector of any vector, divide the x and y-component of that vector with its own magnitude.

$$\hat{a} = \frac{\vec{a}}{|a|} = \left[\frac{a_x}{|a|}, \frac{b_x}{|a|}\right]$$

$\sqrt{0^2+1^2} = \sqrt{1} = 1$

$\hat{a} = \left[\frac{0}{1}, \frac{1}{1}\right] = [0, 1]$

$\hat{b} = \left[\frac{1}{1.4}, \frac{1}{1.4}\right]$ $\sqrt{1^2+1^2} = 1.4$

$\vec{u}$ / $|u|$ mag e / $\hat{u}$ mag · UNIT VECTOR (DIRECTION) = VECTOR ($\vec{u}$)

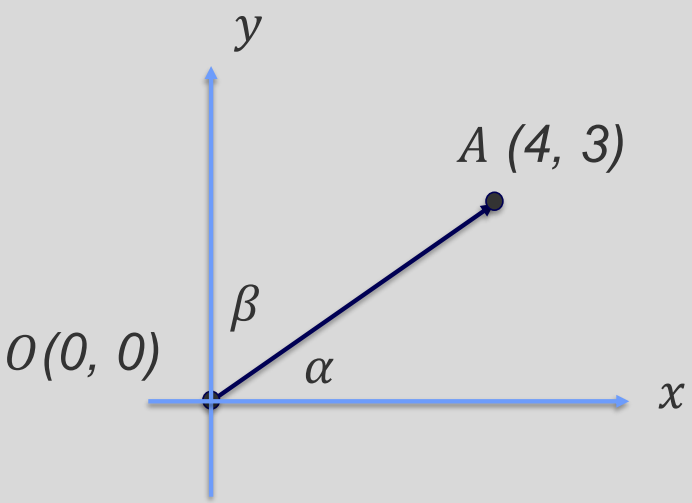
Vector Components – Unit Vector Example

- **Example**: Find the resolution [x, y] for the unit vector $\widehat{OA}$ of vector $\overrightarrow{OA}$ [4, 3].

$|\overrightarrow{OA}| = \sqrt{4^2+3^2} = 5$

$\widehat{OA} = \left[\frac{4}{5}, \frac{3}{5}\right]$

$= [0.8, 0.6)$



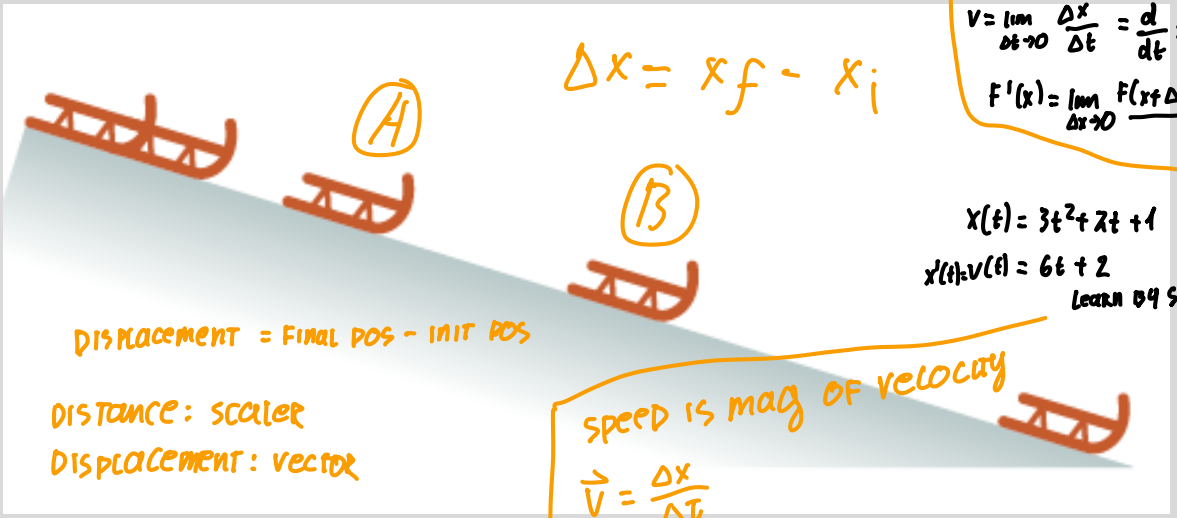
Parameters of Motion

Position, Displacement, Velocity, Acceleration.

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Position vs. Displacement

- Consider a sled sliding downhill, and we have a camera to take a picture of the sled every 1 second.



TENTION LINE

SLOPE $= \frac{52-30}{10-0} = \frac{22}{10}$

$= 2.2$ m/s

$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{d}{dt} x = x'(t)$

$F'(x) = \lim_{\Delta x \to 0} \frac{F(x+\Delta x) - F(x)}{x}$

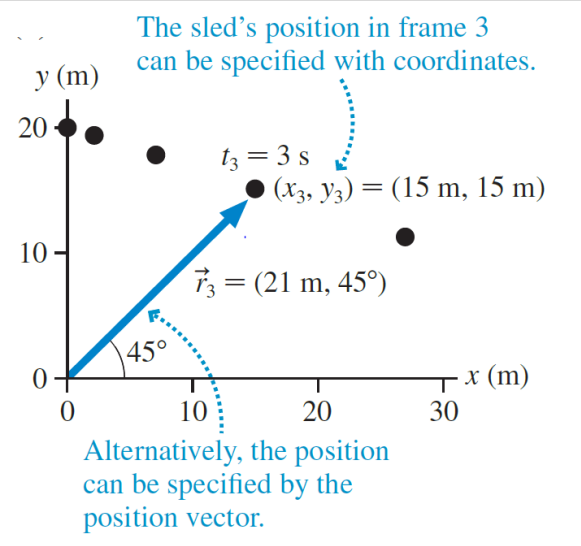
$x(t) = 3t^2 + 2t + 1$

$x'(t) = v(t) = 6t + 2$

LEARN BY SAYING THE WORD

Position vs. Displacement

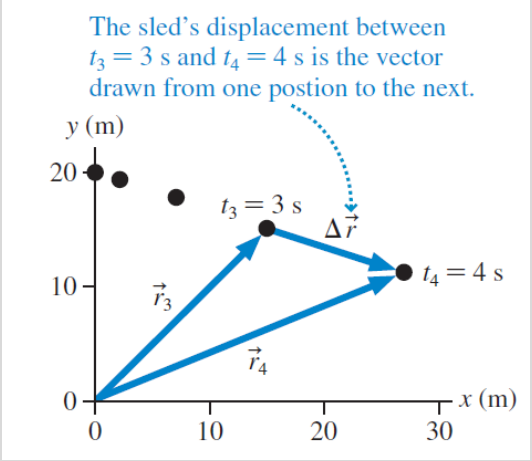
- Thus **the position of the sled** at any given second after its departure can be given by its **coordinate** (x_n, y_n) or the **position vector** $\overrightarrow{r_n}\ [x_n, y_n]$ from our defined origin.



Position vs. Displacement

- The **displacement** of any object is given by its change in position.
- Hence, the **displacement of the sled** is given by the **vector** $\overrightarrow{\Delta r}$ from any point t_{n-1} to t_n. Geometrically, we can calculate $\overrightarrow{\Delta r}$ as:

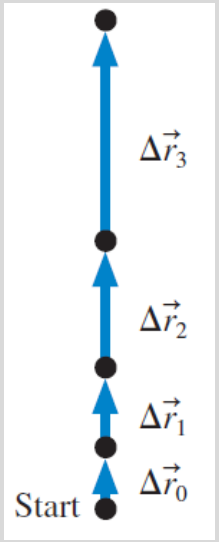
$$\overrightarrow{\Delta r} = \overrightarrow{r_n} - \overrightarrow{r_{n-1}}$$



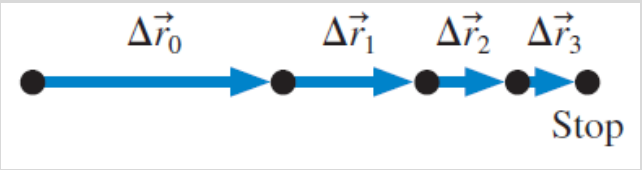
Position vs. Displacement

- The **total displacement** of any object is found by adding all of its individual displacement vectors.

$$\overrightarrow{\Delta r_{total}} = \overrightarrow{\Delta r_0} + \overrightarrow{\Delta r_1} + \overrightarrow{\Delta r_2} + \cdots + \overrightarrow{\Delta r_n}$$



A rocket launching.



A car slowing down to a stop

Displacement vs. Average Speed

- **Average speed** is the quantity that measures an object's fastness or slowness. Unlike displacement, average speed is a scalar value.
- The **average speed** of any object within a time interval is given by:

$$\textit{average speed} = \frac{\textit{distance traveled}}{\textit{time interval spent travelling}} = \frac{d}{\Delta t}$$

- The **distance travelled** is the magnitude of the displacement vector

$$\overrightarrow{(\Delta r)} = \sqrt{\Delta r_x^2 + \Delta r_y^2}$$

Average Speed vs. Velocity

- **Average speed** alone is sufficient for day-to-day activities. However, it is inadequate for most engineering applications.

- The **vector form** of average speed is called **the velocity vector**. Hence, **the average velocity** of an object during the time interval Δt, in which the object undergoes **a displacement** $\Delta\vec{r}$ is given by:

$$\overrightarrow{v_{average}} = \frac{\Delta\vec{r}}{\Delta t}$$

Note: an object's average **velocity vector** $\vec{v}$ points in the same direction as **the displacement vector** $\Delta\vec{r}$. This is the direction of motion.

Velocity vs. Acceleration

- Similar to displacement, **velocity** can change as the object moves. As a vector, there are two ways velocity can change: **magnitude** and **direction**.
- Changes to velocity in either magnitude or direction is called **an acceleration**. Acceleration is defined as the rate of change of velocity with respect to time. Hence, **the average acceleration** of an object during the time interval Δt, in which the object's velocity changes by $\Delta\vec{v}$ is defined as:

$$\overrightarrow{a_{avg}} = \frac{\Delta\vec{v}}{\Delta t}$$

Note: Changes only to the magnitude of the velocity vector (speed) is called **Linear Acceleration** or **Average Acceleration**.

Parameters of Motion – Example

- **Example**: An athlete sprints in a straight line for 50.0 meters in 8 seconds, and then walks slowly back to the starting line in 40.0 seconds. If the "sprint direction" is arbitrarily chosen to be positive, what are (a) the average sprint velocity? (b) the average walk velocity? (c) the average velocity for the entire trip?

